Approaching to the Sample

The Panda robot is equipped with integrated force/torque sensors at each joint. To avoid damaging these sensors - and the load cell mounted beneath the sensor - we must carefully plan the approach motion using the robot's built-in force/torque feedback. The approach motion refers to a slow, controlled movement of the end-effector that begins approximately 1–2 cm above the contact surface and ends once contact with the surface is detected and established. The steps for this motion are as follows:

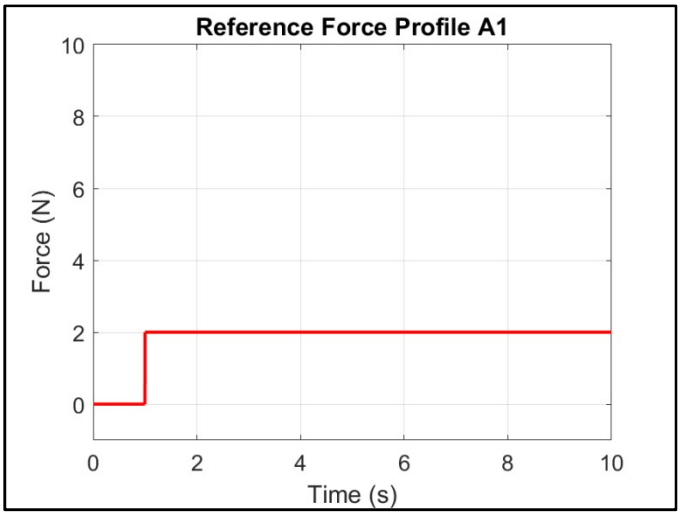
- Position the robot's end-effector (with the 3D-printed contact piece attached, if applicable) directly above the contact surface, maintaining a vertical distance of approximately 1–2 cm from the intended contact point.
- Measure the end-effector's z-axis force at this position and subtract it from the measured force vector to compensate for any offset. In other words, zero the z-axis force reading at 1–2 cm above the contact point by removing the initial bias. We refer to the resulting value as the offset-free force.
- Apply the following approach algorithm: command a controlled downward z-axis velocity of approximately 0.5 mm/s toward the contact point while continuously monitoring the offset-free z-axis force. The algorithm should terminate by setting the z-axis velocity to 0.0 mm/s once the offset-free force exceeds a predefined threshold (e.g., 0.05 N or 0.1 N).
- Once the approach motion terminates at the defined force threshold, we can assume that contact with the sample has been successfully established. It is reasonable to note that this means a small initial force is applied to the system at the moment of contact. However, this minimal preload is necessary to maintain stable contact with the sample. If a small gap remains between the end-effector and the sample, the subsequent motion may cause an unintended impact, which is undesirable and could damage the system or, at minimum, negatively affect the measurement quality.

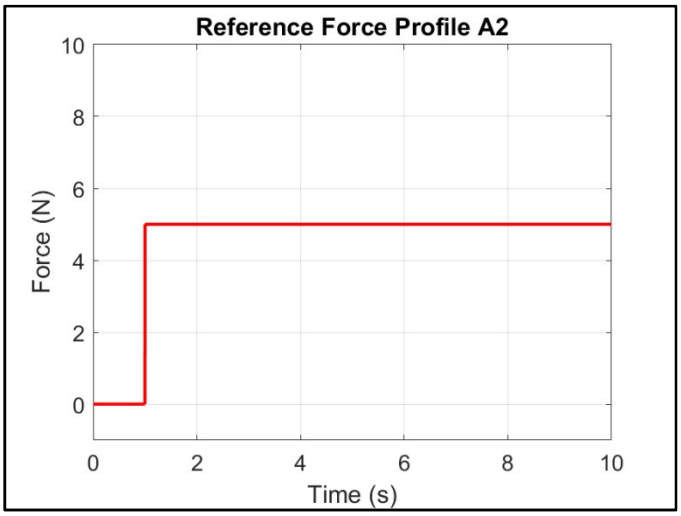
Reference Motion Profiles After Initial Contact

- After establishing stable contact with a small preload force, the next step is to switch the robot's end-effector to force-control mode. This requires verifying that the robot provides a direct force-control interface and confirming its proper operation before proceeding.
- If the robot does not provide a built-in end-effector force-control algorithm, we can instead use a simple end-effector position-control approach supervised by the measured zero-offset z-axis force response. Specifically, we can apply a small-amplitude sinusoidal position reference along the z-axis (e.g., 0.2 mm amplitude) in the compression direction while simultaneously measuring the offset-free z-axis force. This allows us to characterize how the commanded position variation maps to the resulting applied force. The resulting force-position relationship can then be used as an intermediate calibration step to tune the position references needed to achieve the desired force levels.
- Once this mapping is obtained, we can apply the following reference force profiles to the robot's end-effector while simultaneously recording measurements from both the load cell and the readout circuitry of the soft sensor mounted beneath the end-effector:
    - Step Force Profiles
        - Profile A1: A step force reference of 2 N actuating at $t = 1$ s and lasting for 9 s
        - Profile A2: A step force reference of 5 N actuating at $t = 1$ s and lasting for 9 s
        - Profile A3: A step force reference of 8 N actuating at $t = 1$ s and lasting for 9 s
    - Ramp Force Profiles
        - Profile B1: A ramp force reference from 2 N to 8 N in 6 seconds acting at $t = 2$ s
        - Profile B2: A ramp force reference from 2 N to 8 N in 3 seconds acting at $t = 2$ s
        - Profile B3: A ramp force reference from 2 N to 8 N in 1.2 seconds acting at $t = 2$ s
        - Profile B4: A ramp force reference from 2 N to 8 N in 0.6 seconds acting at $t = 2$ s
    - Sinusoidal Force Profiles
        - Profile C1: A sinus force ref of $F = 5 + 3\sin(t - 1)$ N for 8 seconds acting at $t = 1$ s
        - Profile C2: A sinus force ref of $F = 5 + 3\sin(2t - 2)$ N for 8 seconds acting at $t = 1$ s
        - Profile C3: A sinus force ref of $F = 5 + 3\sin(3t - 3)$ N for 8 seconds acting at $t = 1$ s
        - Profile C4: A sinus force ref of $F = 5 + 3\sin(4t - 4)$ N for 8 seconds acting at $t = 1$ s
        - Profile C5: A sinus force ref of $F = 5 + 3\sin(5t - 5)$ N for 8 seconds acting at $t = 1$ s

- Sinusoidal Sweep Force Profile
  - Profile D1: A sweep ref of $F = 5 + 3\sin((t-1)^2)$ N for 8 seconds acting at $t = 1$ s
  - Profile D2: A sweep ref of $F = 5 + 3\sin((2t-2)^2)$ N for 8 seconds acting at $t = 1$ s
  - Profile D3: A sweep ref of $F = 5 + 3\sin((3t-3)^2)$ N for 8 seconds acting at $t = 1$ s

These motion profiles are shown on the graphs in the following pages.


Reference Force Profile A1
Force (N)
Time (s)
0
2
4
6
8
10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Force (N)
Time (s)
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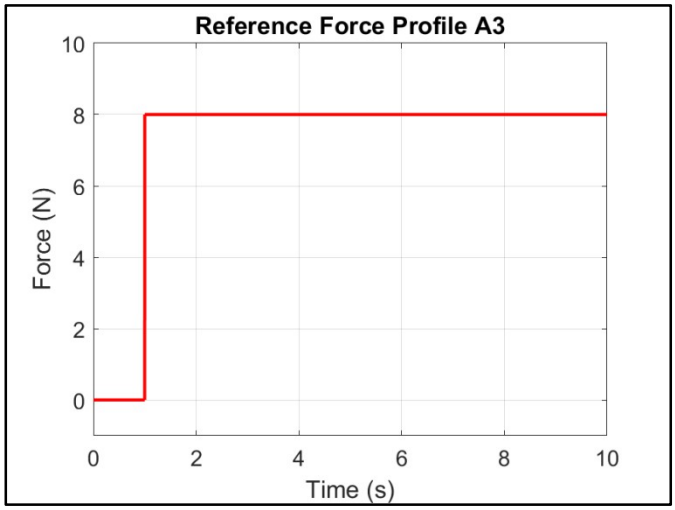
2
4
6
8
10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Force (N)
Time (s)
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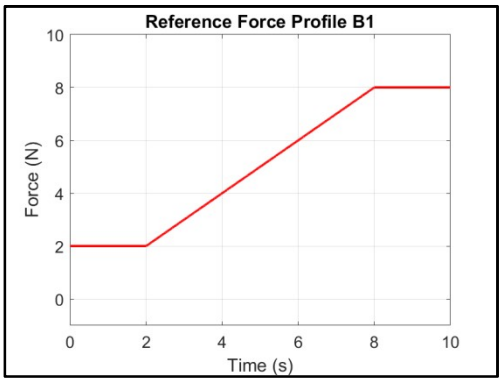
8
10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Force (N)
Time (s)
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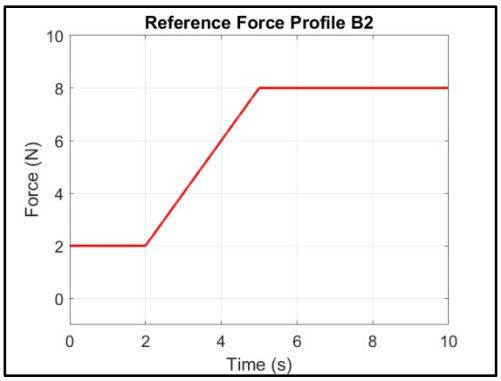
2
4
6
8
10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Force (N)
Time (s)
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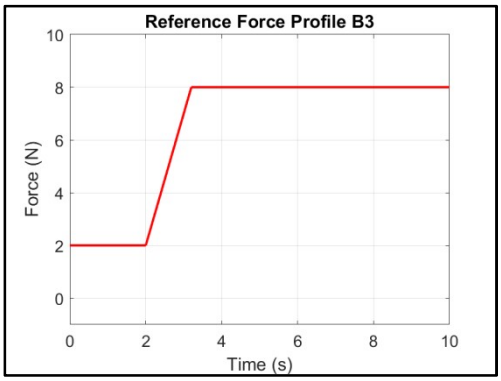
2
4
6
8
10


Reference Force Profile B3
Force (N)
Time (s)
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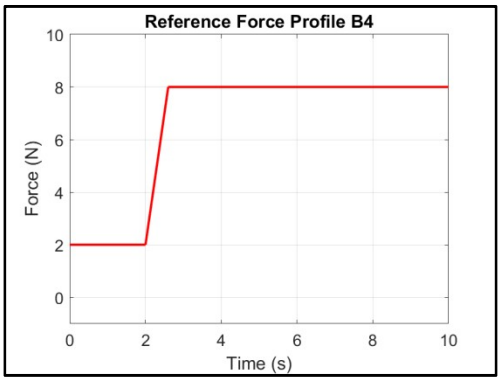
2
4
6
8
10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Force (N)
Time (s)
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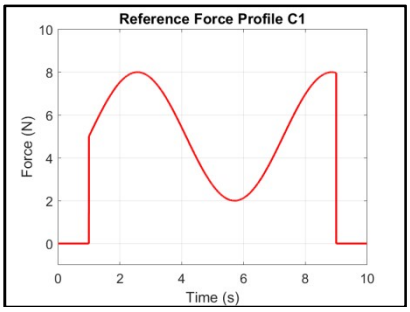
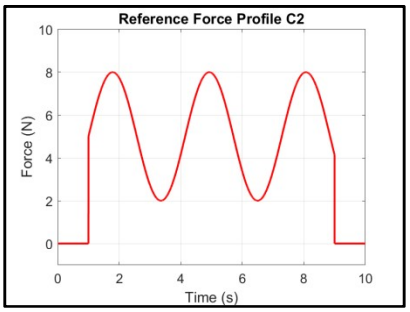
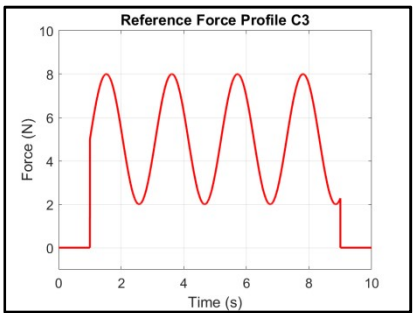
2
4
6
8
10


Reference Force Profile C1
Force (N)
Time (s)
0
2
4
6
8
10


Reference Force Profile C2
Force (N)
Time (s)
0
2
4
6
8
10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Force (N)
Time (s)
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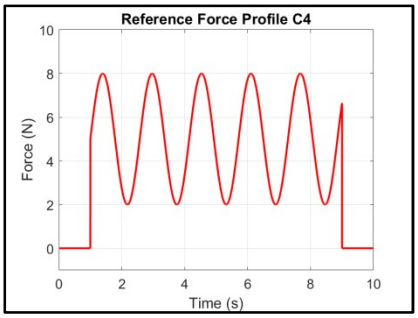
2
4
6
8
10


Reference Force Profile C4
Force (N)
Time (s)
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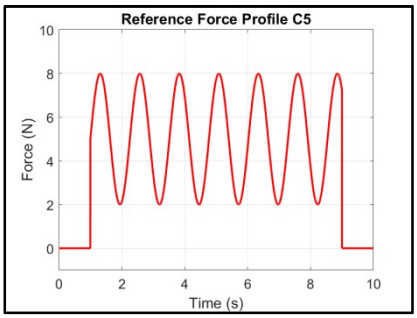
2
4
6
8
10


Reference Force Profile C5
Force (N)
Time (s)
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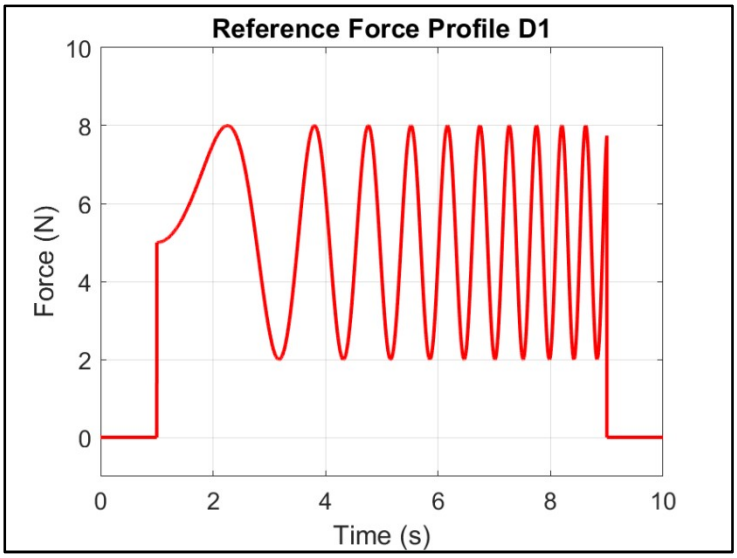
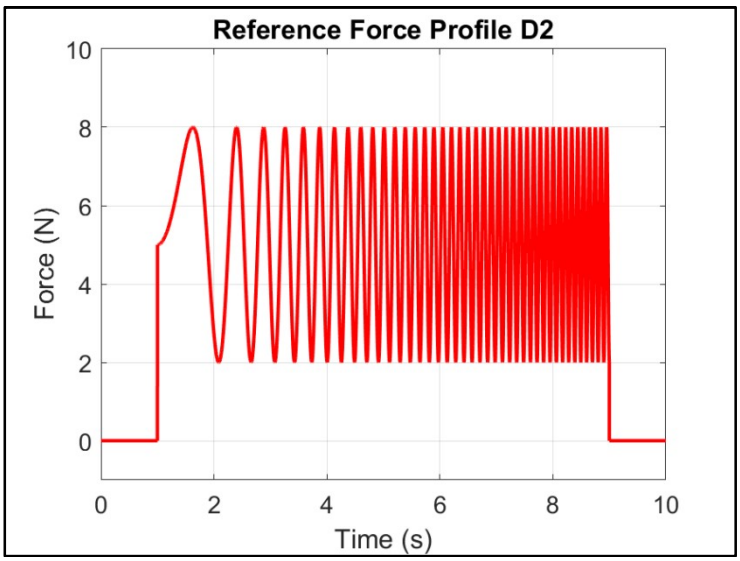
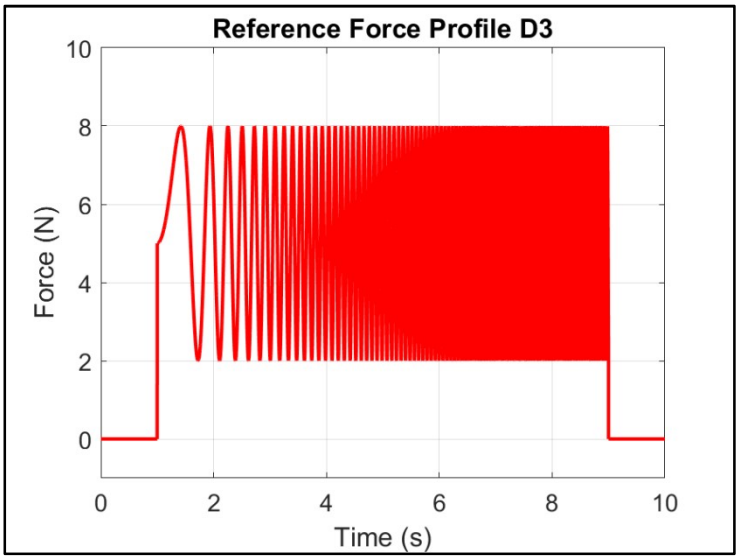
2
4
6
8
10


Reference Force Profile D1
Force (N)
Time (s)
10
8
6
4
2
0
0
2
4
6
8
10


Reference Force Profile D2
Force (N)
Time (s)
10
8
6
4
2
0
0
2
4
6
8
10


Reference Force Profile D3
Force (N)
Time (s)
10
8
6
4
2
0
0
2
4
6
8
10